

On the Harborth Conjecture

Part II

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Abstract

Harborth's conjecture states that every planar graph has a crossing-free straight-line drawing in which every edge has an integer length. Kleber's strengthening asks for the vertices themselves to have integer coordinates. In this series of papers, we make progress towards settling these conjectures. We reduce Kleber's conjecture to local rational-distance statements for special polygons with at most five vertices. The triangle case is known from the results of Almering and Berry. In Part I, we proved the existence of a rational-distance point on an interior integer diagonal in all the cases for non-degenerate quadrilaterals. In this Part II paper, we start working on non-degenerate pentagons. In the upcoming papers, we focus on more non-degenerate pentagons and then degenerate quadrilaterals and degenerate pentagons.

Keywords: algebraic number theory, algebraic geometry, planar graphs, integral drawings, elliptic curves, rational distances, rational points.

1 Introduction

Harborth's conjecture asks whether every finite planar graph admits a crossing-free straight-line drawing in which every edge has integer length [1]. Kleber asked for the stronger form in which the vertices themselves have integer coordinates [10]. A standard way to attack these problems is local and inductive. One removes a low-degree vertex from a triangulated planar graph, draws the smaller graph, and then tries to put the deleted vertex back inside the face which has been created. The local face has at most five boundary vertices. Thus the insertion problem naturally breaks into the triangle, quadrilateral, and pentagon cases.

The triangle case is covered by classical rational-distance results of Almering and Berry. The quadrilateral case is the first genuinely new local step: one seeks a rational-distance point on an interior diagonal of a rational or integer quadrilateral. The preceding quadrilateral part of this project proves that insertion statement for simple nondegenerate quadrilaterals. The present paper is the next stage. It treats the pentagon local face, where visibility is no longer automatic from being inside the polygon and where the arithmetic curves attached to a diagonal can have genus 5.

Accordingly, this paper should be read as Part 2 of the same program. It does not claim a uniform solution of every pentagon case. Instead, it organizes the pentagon problem into a finite and checkable set of geometric, arithmetic, and certificate branches. The point of the organization is to make the remaining work precise: to prove the pentagon insertion theorem uniformly, one must produce a positive certificate on at least one distinguished diagonal in every hard branch.

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1.1 Pentagon insertion problem

We study a restricted rational distance problem for simple integer pentagons. Let

$$Q = (O, A, X, C, B)$$

be a simple nondegenerate pentagon in cyclic order. The standing hypotheses are that the vertices have integer coordinates, the five side lengths are integers, and the two diagonals

$$OX, \quad OC$$

are interior diagonals of integer length. Thus the two distinguished diagonals share the vertex O .

The problem is to find a rational point P , lying in the relative interior of one of these two diagonals, such that

$$P \in \mathcal{K}(Q)$$

and

$$PO, PA, PX, PC, PB \in \mathbb{Q}.$$

Here $\mathcal{K}(Q)$ denotes the kernel of the polygon: the set of points of Q from which every point of Q is visible by a segment contained in Q . The kernel condition is therefore stronger than asking P to see only the five vertices. It requires P to be a globally visible point of the pentagonal region.

Equivalently, the goal is to certify at least one of the two diagonal problems (Q, OX) and (Q, OC) . If L is one of the two distinguished diagonals, the relevant geometric set is the visible interval

$$J_L = (\text{relint } L) \cap \mathcal{K}(Q).$$

The diagonal L is certified when J_L contains a rational point whose distances to all five vertices are rational. The pentagon is certified when at least one distinguished diagonal is certified. In the Harborth/Kleber induction, such a certified point is the candidate position for the deleted vertex after a final common scaling of rational edge lengths.

1.2 Why the pentagon case is harder

The pentagon case has two extra difficulties beyond the quadrilateral case. First, the candidate point is not allowed to range over the whole plane; it must lie on one of two prescribed diagonals. This converts the metric problem into a one-dimensional arithmetic problem. After normalizing a chosen diagonal to the x -axis, the distances to the two endpoints are rational for rational x , while each off-line vertex $Y_i = (a_i, b_i)$ contributes a condition of the form

$$(x - a_i)^2 + b_i^2 \in \mathbb{Q}^\square.$$

Thus the arithmetic is controlled by fiber products of elementary conics, which have genus 0, 1, or 5 according to the number of distinct mirror classes of the off-line vertices.

Second, the point must lie in the kernel of the pentagon. For quadrilaterals, a point on a suitable interior diagonal sees all four vertices in the relevant local argument. For pentagons, visibility from all points of the polygon is subtler: the point must lie in $\mathcal{K}(Q)$, the set from which the entire pentagonal region is visible. Thus visibility is a geometric constraint, while rational distance is an arithmetic constraint. Mixing these two kinds of data in a single case split quickly obscures

which part of a proof is doing the real work. Convexity, concavity, kernel shape, and niceness of a diagonal are all useful geometric descriptors, but none of them determines the arithmetic curve attached to a diagonal. Conversely, an arithmetic theorem about rational points on a line is useful for the pentagon only after those rational points are shown to meet the visible interval.

The purpose of the three-layer taxonomy is to separate these issues. The geometry layer computes J_L . The arithmetic layer identifies the curve whose rational points encode the distance conditions. The certificate layer proves that the relevant rational points actually occur in the visible interval. This makes clear which cases are already proved, which ones are reduced to standard computations, and which hard branches still require new certificates.

1.3 Related work

The graph-drawing motivation comes from Harborth’s conjecture [1] and Kleber’s coordinate strengthening [10]. The local insertion viewpoint reduces the relevant metric questions to rational distance problems in small polygons. The geometric input for the pentagon stage is the standard theory of polygon kernels. For a simple polygon, the kernel is the intersection of the closed interior half-planes determined by its oriented sides; see O’Rourke [14, Chapter 2]. Here this theory is used in a one-dimensional form. We intersect the kernel with each distinguished diagonal and obtain an interval J_L with rational endpoints.

The main density input in the nonexceptional genus-one case is Love’s detour theorem for rational configuration problems [11]. In the endpoint-complete two-class branch, the two distance conditions can be written as a rational detour equation

$$s\alpha_1 + \alpha_2 = r.$$

Away from Love’s exceptional locus, a known rational detour point is non-torsion on the associated elliptic curve, and its multiples give density on the relevant real component. This is the mechanism which turns endpoint control into infinitely many certified points in every nonempty open visible interval.

The exceptional and one-endpoint genus-one branches naturally lead to elliptic curves. The certificate layer records the elliptic data needed in those cases: a non-torsion point meeting the correct real component, or a complete rank-zero and torsion computation. The general background for these computations is the arithmetic of elliptic curves, including rank, torsion, and explicit algorithms; see Silverman [15], Mazur [12], and Cremona [5].

When three distinct mirror classes occur, the line problem has genus 5. Faltings’ theorem gives finiteness of rational points on curves of genus at least 2 [6], but a proof for a concrete pentagon requires effective information. The certificate layer is designed to record such information, for example through descent, Chabauty–Coleman methods, Mordell–Weil sieves, and quotient decompositions; see Chabauty [3], Coleman [4], Bruin–Stoll [2], Stoll [16], and Kani–Rosen [9]. The present paper does not hide these computations inside the taxonomy: a genus 5 branch is positive only when the recorded certificate produces a rational point whose diagonal coordinate lies in the visible interval.

1.4 Proof architecture and main output

A point of the plane is called rational if both of its coordinates are in $\mathbb{Q}$. A segment is called integer if its Euclidean length lies in $\mathbb{Z}_{>0}$. These conventions are used throughout the paper. The formal definitions begin in the next section; the following paragraph summarizes the proof architecture.

The basic bookkeeping object is a diagonal profile

$$\Pi(Q, L) = (\text{Geom}(Q, L), \text{Arith}(Q, L), \text{Cert}(Q, L)), \quad L \in \{OX, OC\}.$$

The first entry is geometric: it records the visible interval J_L . The second entry is arithmetic: it records the mirror classes of the off-line vertices, the endpoint-control data, the detour parameters in genus one, and the genus of the associated curve. The third entry is the certificate: it records the theorem, explicit point, elliptic computation, torsion calculation, local obstruction, or genus 5 computation which proves that the arithmetic points do, or do not, hit J_L .

A positive certificate on either distinguished diagonal proves the desired existence statement for the pentagon. A negative certificate only rules out that particular diagonal and sends the proof to the other one. This distinction is essential in the hard cases: convexity or niceness may give full visibility, but a rational visible point still requires an arithmetic certificate; conversely, an arithmetic curve may have many rational points which are irrelevant unless their x -coordinates lie in the interval allowed by the kernel.

The main output is therefore a certified case division. One-class lines are proved by density of the elementary Pythagorean-slope set, although they are vacuous for simple nondegenerate pentagons. Endpoint-complete nonexceptional two-class lines are proved using Love's detour density theorem and give infinitely many certified points in every nonempty open visible interval. Exceptional genus-one lines, one-endpoint genus-one lines, and genuine genus 5 lines are isolated as the hard branches. For these branches the paper states exactly what kind of positive certificate finishes the diagonal and what kind of negative computation only rules that diagonal out. This is the promised pentagon-stage progress toward the Harborth program: the remaining obstruction is no longer a vague pentagon case, but a precise list of arithmetic certificates to be supplied.

2 The square branch and the four-distance problem

There is one elementary square branch which should be separated from the main taxonomy. It explains why the results below should not be read as a solution of the classical four-distance problem for the unit square, and also why the square branch is not automatically solved by the existence of the fifth vertex O .

Proposition 2.1 (Four vertices forming a square). *Under the standing hypotheses, four vertices of Q cannot form a square which contains O . Hence, if four vertices of Q form a square, the only possible set is*

$$\{A, X, C, B\}.$$

In that case the four vertices A, X, C, B occur in square cyclic order, the square side length is an integer, and O is an integer-distance point from all four vertices of this square. After scaling the square to unit side length, O gives a point at rational distance from all four vertices of the unit square.

Proof. Suppose first that a square formed by four vertices contains O . The three distances from O to the other three vertices of the square are chosen from

$$OA, OX, OC, OB,$$

all of which are integers under the standing hypotheses. But from one vertex of a nondegenerate square, the distances to the other three vertices are

$$s, \quad s, \quad s\sqrt{2},$$

where $s > 0$ is the square side length. Thus both s and $s\sqrt{2}$ would be integers, which is impossible.

Therefore a four-vertex square, if it exists, must be $\{A, X, C, B\}$. The three pentagon sides

$$AX, \quad XC, \quad CB$$

are integers. In a square, a segment joining two vertices is either a side of length s or a diagonal of length $s\sqrt{2}$. A path through the four square vertices cannot have all three successive edges diagonal. Thus if one of AX, XC, CB were a square diagonal, another would be a square side, forcing both s and $s\sqrt{2}$ to be integers. This is again impossible. Hence AX, XC, CB are all square sides, so A, X, C, B occur in square cyclic order and $s \in \mathbb{Z}_{>0}$.

Finally,

$$OA, \quad OX, \quad OC, \quad OB \in \mathbb{Z}$$

by the side and distinguished-diagonal hypotheses. Scaling the square by $1/s$ sends it to the unit square and scales these four distances by the rational factor $1/s$. Hence the image of O is at rational distance from all four vertices of the unit square. $\square$

The four-distance problem asks whether there exists a point in the plane at rational distance from all four vertices of the unit square; see Guy [7] and McCloskey [13]. Several special cases are known, but the general problem remains open [8]. To avoid ambiguity about the word “conjecture,” let $(\text{FD} * +)$ denote the positive assertion that such a point exists, and let $(\text{FD} * -)$ denote its negation, which is the commonly expected outcome.

Proposition 2.1 shows that the square branch in the present pentagon problem can occur only under $(\text{FD} * +)$, and only after the additional pentagon conditions of simplicity, nondegeneracy, and interiority of OX and OC are imposed. Thus this paper is not solving the famous four-distance problem. If $(\text{FD} * -)$ is true, then no pentagon in the present class has four vertices forming a square. If (FD_+) is true, then the square branch cannot be dismissed on four-distance grounds; every such instance which also satisfies the pentagon hypotheses must be solved by the three-layer taxonomy below, with its own geometry, arithmetic, and certificate data.

Proposition 2.2 (The square branch is a certificate problem). *Assume that A, X, C, B form a square. Then neither endpoint X nor endpoint C gives the required rational-distance point, and O is not an admissible point because the problem requires the relative interior of a distinguished diagonal.*

Moreover, in the terminology of Definition 6.7, both distinguished diagonals are remote-vertex genus 5 lines, namely Type IIIb. On OX , the remote square vertex is B ; on OC , the remote square vertex is A . Consequently the existence of a rational point in

$$\text{relint}(OX) \cup \text{relint}(OC)$$

is not automatic from the square hypothesis. Within this taxonomy, such a point is certified by producing a positive Type IIIb certificate on at least one of the two distinguished diagonals.

Proof. Let $s \in \mathbb{Z}_{>0}$ be the side length of the square A, X, C, B , in that cyclic order. At the endpoint X , the distance to the opposite square vertex is

$$XB = s\sqrt{2} \notin \mathbb{Q}.$$

At the endpoint C , the distance to the opposite square vertex is

$$CA = s\sqrt{2} \notin \mathbb{Q}.$$

Thus neither square endpoint can be used as a rational-distance point. The point O is at integer distance from all four square vertices by Proposition 2.1, but O is an endpoint of both distinguished diagonals, not a point of their relative interiors.

Consider the diagonal OX . The three off-line vertices are A, C, B . The far endpoint X controls the distances to A and C , since

$$XA = XC = s,$$

but it does not control the distance to B , since $XB = s\sqrt{2}$. Hence the class of B is remote from the far endpoint. The common endpoint O controls all three off-line vertices, because

$$OA, OC, OB \in \mathbb{Z}.$$

It remains to check that the three off-line vertices determine three distinct mirror classes with respect to OX . The vertex B cannot be mirror-equivalent to A or C , because reflection across OX preserves distance from X , whereas $XB = s\sqrt{2}$ and

$$XA = XC = s.$$

If A and C were mirror-equivalent with respect to OX , then OX would be an angle bisector of the right angle $\angle AXC$. Hence the line OX would be parallel to one of the two square diagonals. Since the square has integer vertices, such a line has rational direction vector parallel to a square diagonal, and every nonzero integer segment in that direction has length a rational multiple of $s\sqrt{2}$. This cannot be an integer. This contradicts the standing hypothesis that OX has integer length. Therefore the OX -line has three mirror classes and exactly one remote class, namely the class of B . It is Type IIIb.

The proof for OC is the same with the roles rotated. The far endpoint C controls X and B , because

$$CX = CB = s,$$

but not A , because $CA = s\sqrt{2}$. The vertex A cannot be mirror-equivalent to X or B , since the distances from C differ. If X and B were mirror-equivalent with respect to OC , then OC would be parallel to a square diagonal, contradicting the integer length of OC by the same argument as above. Thus OC is also Type IIIb, with remote class represented by A . Again the common endpoint O controls all three off-line vertices, because

$$OA, OC, OB \in \mathbb{Z}.$$

By the certificate criterion for Type III branches in Proposition 7.10, a positive genus 5 certificate whose x -coordinate lies in the visible interval certifies a rational point on one of the two interior distinguished diagonals. A negative resolution of one diagonal only rules out that diagonal and does not certify the pentagon. Thus the square hypothesis supplies the four-distance endpoint O , but it does not by itself supply an admissible interior point. $\square$

3 Setup and guiding principle

Throughout, a point of the plane is called *rational* if both coordinates lie in $\mathbb{Q}$. A segment is called *integer* if its Euclidean length lies in $\mathbb{Z}_{>0}$.

Let

$$Q = (O, A, X, C, B)$$

be a simple pentagon, written in cyclic order, and let the same letter Q denote its closed polygonal region. We assume that Q is *nondegenerate*, meaning that no three of its vertices are collinear. We also assume that all vertices have integer coordinates, all five side lengths are integer, and that the two diagonals

$$OX, \quad OC$$

are interior integer diagonals sharing the common vertex O . Here *interior* means that the relative interior of the diagonal is contained in the interior of the polygonal region Q .

The problem is to find a rational point

$$P \in \text{relint}(OX) \cup \text{relint}(OC)$$

such that

$$P \in \mathcal{K}(Q)$$

and the five distances from P to the vertices of Q are rational. The kernel condition is stronger than visibility from the five vertices: it ensures that P sees every point of the pentagon.

The classification will be made for one distinguished diagonal at a time. Thus the basic object is the pair

$$(Q, L), \quad L \in \{OX, OC\}.$$

As made precise below, the pentagon is certified once at least one of the two diagonal problems (Q, OX) or (Q, OC) is certified.

Thus the global labels ‘convex’, ‘concave’, ‘nice’, and ‘non-nice’ are not the primary taxonomy. They are used in the geometry layer, where they help determine the visible intervals on the two diagonals.

Definition 3.1 (Kernel and nice diagonals). The kernel $\mathcal{K}(Q)$ is the set of points $P \in Q$ such that

$$[P, Y] \subset Q$$

for every $Y \in Q$. Equivalently, for a simple polygon, $\mathcal{K}(Q)$ is the intersection of the five closed interior half-planes determined by the oriented sides of Q ; see, for example, O’Rourke [14, Chapter 2]. This half-plane description is the one used below to compute visible intervals.

A distinguished diagonal $L \in \{OX, OC\}$ is called *nice* if the whole closed diagonal segment lies in the kernel:

$$L \subset \mathcal{K}(Q).$$

It is called *non-nice* if this containment fails.

Definition 3.2 (Visible interval and certified diagonal). Let $L \in \{OX, OC\}$. The visible interval on L is

$$J_L = (\text{relint } L) \cap \mathcal{K}(Q)$$

viewed in a rational coordinate along the line containing L . More explicitly, if $L = UV$ has integer endpoints and integer length D_L , we use the coordinate $t \in [0, D_L]$ defined by

$$P = U + \frac{t}{D_L}(V - U).$$

Then rational values of t correspond to rational points of L , and conversely.

The diagonal problem (Q, L) is called *certified* if one has a proof that J_L contains a rational point whose distances from all five vertices of Q are rational. The pentagon Q is called *certified* if either (Q, OX) or (Q, OC) is certified.

Definition 3.3 (Three-layer profile). For a distinguished diagonal L , its three-layer profile is the triple

$$\Pi(Q, L) = (\text{Geom}(Q, L), \text{Arith}(Q, L), \text{Cert}(Q, L)).$$

The three entries mean:

1. $\text{Geom}(Q, L)$ records the interval J_L ;
2. $\text{Arith}(Q, L)$ records the mirror classes of the off-line vertices, the endpoint-control data, and the genus of the associated line curve;
3. $\text{Cert}(Q, L)$ records the theorem, explicit point, elliptic computation, torsion computation, or genus 5 computation that proves the arithmetic rational points meet J_L .

4 Normalization

The lemma below justifies the following normal form. After choosing one of the two distinguished diagonals to be the horizontal diagonal, we use the notation

$$O = (0, 0), \quad X = (D, 0), \quad D \in \mathbb{Z}_{>0},$$

and

$$A = (a, b), \quad C = (m, n), \quad B = (p, q),$$

with integer coordinates and

$$E = OC = \sqrt{m^2 + n^2} \in \mathbb{Z}_{>0}.$$

Because Q is nondegenerate, the vertices O, X, C are not collinear; therefore $n \neq 0$. Reflecting in the x -axis, if necessary, preserves all metric, rationality, and visibility conditions, so we assume

$$n > 0.$$

When a nice diagonal exists, the preferred convention is to choose a nice distinguished diagonal as the horizontal one. If no nice diagonal exists, the same normalization can still be applied to either distinguished integer diagonal; only the full-visibility shortcut is then unavailable.

Lemma 4.1 (Integer diagonal normalization). *Let Q have integer vertices, and let UV be a diagonal of integer length*

$$\ell = UV \in \mathbb{Z}_{>0}.$$

After translating, applying a rational orthogonal change of coordinates, and multiplying all coordinates by ℓ , the diagonal UV becomes

$$(0, 0)(\ell^2, 0).$$

All vertices remain integral, all side lengths remain integral after the common scaling, rational points correspond to rational points, rational distances correspond to rational distances, and kernel membership is preserved. If UV is nice, then the transformed diagonal is nice.

Proof. Translate so that $U = (0, 0)$, and write $V = (m, n)$ with

$$\ell = \sqrt{m^2 + n^2} \in \mathbb{Z}_{>0}.$$

Since U is an integer vertex, this translation preserves integrality of all vertex coordinates. The orthogonal map

$$R(x, y) = \left(\frac{mx + ny}{\ell}, \frac{-nx + my}{\ell} \right)$$

has rational coefficients and sends V to $(\ell, 0)$. Multiplying by ℓ gives

$$T(x, y) = \ell R(x, y) = (mx + ny, -nx + my),$$

which has integral coefficients and sends V to $(\ell^2, 0)$. Thus integer vertices are sent to integer points.

The map T is a similarity with scale factor ℓ . Hence every side length is multiplied by ℓ , so integer side lengths remain integer. A distance is rational before applying T if and only if its image distance is rational, because the scale factor ℓ is a nonzero integer. The inverse map has rational coefficients, explicitly

$$T^{-1}(x, y) = \frac{1}{\ell^2}(mx - ny, nx + my),$$

so rational points correspond in both directions.

Similarities preserve segments and incidences. Therefore, for every point $P \in Q$,

$$P \in \mathcal{K}(Q) \iff T(P) \in \mathcal{K}(T(Q)).$$

Equivalently,

$$T(\mathcal{K}(Q)) = \mathcal{K}(T(Q)).$$

Thus the kernel condition for candidate points is unchanged by the normalization. If, in addition, $UV \subset \mathcal{K}(Q)$, then

$$T(UV) \subset \mathcal{K}(T(Q)),$$

so the transformed diagonal is nice. □

Definition 4.2 (Nice-first convention). If at least one distinguished diagonal is nice, choose a nice distinguished diagonal as the normalized horizontal diagonal OX . If the originally labelled diagonal OC is chosen, relabel the pentagon in the reverse cyclic order

$$(O, B, C, X, A)$$

before applying Lemma 4.1; then the chosen original diagonal OC becomes the new horizontal diagonal OX , and the normalized pentagon is again denoted

$$(O, A, X, C, B).$$

If exactly one distinguished diagonal is nice, this choice is forced. If neither distinguished diagonal is nice, either distinguished diagonal may be normalized as OX , but the actual visible interval on that non-nice diagonal must be computed before applying any arithmetic theorem.

5 Layer 1: geometry

The geometry layer answers only one question for each distinguished diagonal L : what is the visible interval J_L ?

Definition 5.1 (Geometric visibility type). Let L be a distinguished diagonal of length D_L , and use a rational coordinate $t \in [0, D_L]$ along it. The following equalities and open intervals are understood in this coordinate. The diagonal has:

1. *full visibility* if $J_L = (0, D_L)$;
2. *usable partial visibility* if J_L is not full but contains a nonempty open interval;
3. *degenerate visibility* if J_L is empty or contains no nonempty open interval.

This is the preferred geometric classification. Convexity, concavity, and niceness are methods for determining which of these three alternatives holds.

Definition 5.2 (Convexity and concavity). The pentagon Q is called *convex* if its closed polygonal region is a convex subset of the plane. It is called *strictly convex* if it is convex and every interior angle is strictly smaller than π . In particular, a strictly convex pentagon has no three consecutive collinear vertices. A simple pentagon that is not convex is called *concave*.

Lemma 5.3 (Rational kernel intervals). Assume $Q = (O, A, X, C, B)$ is in the normalized coordinates of Section 4:

$$O = (0, 0), \quad X = (D, 0), \quad C = (m, n),$$

with rational vertices and rational distinguished diagonal lengths

$$D = OX, \quad E = OC = \sqrt{m^2 + n^2}.$$

Then

$$\mathcal{K}(Q) \cap OX \quad \text{and} \quad \mathcal{K}(Q) \cap OC$$

are possibly empty closed intervals in their respective diagonal lines. In the coordinates $x \in [0, D]$ on OX and $u \in [0, E]$ on OC , their endpoints are rational.

Proof. The kernel is the intersection of the five closed interior half-planes defined by the sides of Q . Since the vertices are rational, each supporting line has a rational equation. Restricting these five rational linear inequalities to the rational line OX gives rational linear inequalities in x . Their intersection with $0 \leq x \leq D$ is a closed interval with rational endpoints, possibly empty.

For OC , use the coordinate

$$P = \frac{u}{E}C.$$

Since $m, n, E \in \mathbb{Q}$ and $E \neq 0$, this is a rational parametrization of the line OC . Substituting it into the same five half-plane inequalities gives rational linear inequalities in u . Their intersection with $0 \leq u \leq E$ is again a possibly empty closed interval with rational endpoints. $\square$

Lemma 5.4 (The common vertex is visible). *Let $Q = (O, A, X, C, B)$ be simple and suppose OX and OC are interior diagonals. Then*

$$O \in \mathcal{K}(Q).$$

Consequently the closed intervals $\mathcal{K}(Q) \cap OX$ and $\mathcal{K}(Q) \cap OC$ are nonempty.

Proof. Because OX and OC are interior diagonals in the cyclic pentagon (O, A, X, C, B) , cutting Q along them gives the fan triangulation

$$Q = \triangle OAX \cup \triangle OXC \cup \triangle OCB.$$

The intersections of these three closed triangles are only common edges, common vertices, or empty sets. Every point $Y \in Q$ lies in one of these closed triangles, and every one of the triangles contains O . Since each triangle is convex, the segment $[O, Y]$ lies in that triangle and hence in Q . Thus O sees every point of Q . $\square$

Proposition 5.5 (Convex full-visibility shortcut). *If Q is convex, then*

$$\mathcal{K}(Q) = Q.$$

If, moreover, OX and OC are interior diagonals, then

$$J_X = (0, D), \quad J_C = (0, E).$$

Under the standing nondegeneracy hypothesis, convex and strictly convex coincide.

Proof. If Q is convex, the segment between any two points of Q lies in Q . Hence every point of Q sees every point of Q , so $\mathcal{K}(Q) = Q$. Since the two diagonals are assumed interior, their relative interiors lie inside Q , giving the displayed intervals. If a convex pentagon in this paper were not strictly convex, it would have a flat angle, hence three collinear vertices, contrary to nondegeneracy. $\square$

Proposition 5.6 (Nice-diagonal full-visibility shortcut). *If a distinguished diagonal L is nice, then*

$$J_L = \text{relint } L.$$

Thus a nice diagonal has full visibility, meaning $J_L = (0, D_L)$ in its diagonal coordinate. If Q is nice, meaning both distinguished diagonals are nice, then

$$J_X = (0, D), \quad J_C = (0, E).$$

Proof. If $L \subset \mathcal{K}(Q)$, then

$$\mathcal{K}(Q) \cap L = L.$$

Intersecting with the relative interior of L gives the claim. Applying this to both distinguished diagonals gives the final statement. $\square$

Proposition 5.7 (Geometry-layer results already proved). *For each distinguished diagonal L , the geometry layer proves the following visibility statements.*

1. *The set $J_L = (\text{relint } L) \cap \mathcal{K}(Q)$ is an interval with rational endpoints in the diagonal coordinate, possibly empty.*
2. *The common endpoint O belongs to $\mathcal{K}(Q)$.*
3. *If Q is convex, then both distinguished diagonals have full visibility.*
4. *If L is nice, then L has full visibility.*

These are exactly geometry-layer conclusions. They do not, by themselves, assert the existence of a rational point at rational distance from all five vertices.

Proof. The rational-interval assertion is Lemma 5.3, after intersecting the closed interval $\mathcal{K}(Q) \cap L$ with $\text{relint } L$. The assertion $O \in \mathcal{K}(Q)$ is Lemma 5.4. The convex full-visibility assertion is Proposition 5.5, and the nice-diagonal assertion is Proposition 5.6.

All four statements concern membership in the kernel and the shape of visible intervals. The rational-distance conditions are the separate arithmetic conditions introduced in Section 4. Therefore a geometry-layer result certifies a pentagon only after it is combined with an arithmetic theorem or a positive certificate from the certificate layer. $\square$

Remark 5.8 (Why convexity is not the main taxonomy). Convexity is a powerful shortcut in the geometry layer because it implies $\mathcal{K}(Q) = Q$. It does not solve the arithmetic layer. A convex pentagon may still have an exceptional genus-one line, a one-endpoint genus-one line, or a genuine genus 5 line.

Remark 5.9 (Why kernel shape is not the main taxonomy). The full two-dimensional shape of $\mathcal{K}(Q)$ is usually more information than the proof needs. The arithmetic theorems use only the one-dimensional interval J_L on a chosen diagonal. Therefore the clean geometry output is the pair of intervals J_X, J_C , not a case split by all possible kernel polygons.

6 Layer 2: arithmetic

The arithmetic layer classifies the rational-distance problem on a fixed usable diagonal L . Normalize L to the x -axis by Lemma 4.1, with endpoints

$$U = (0, 0), \quad V = (D_L, 0).$$

Write the three off-line vertices as

$$Y_i = (a_i, b_i), \quad b_i \neq 0, \quad i = 1, 2, 3.$$

Here $a_i, b_i \in \mathbb{Q}$. The nonzero height condition follows from nondegeneracy: an off-line vertex on the line through U, V would give three collinear vertices of Q . For $P = (x, 0)$, the endpoint distances are rational whenever

$$x \in \mathbb{Q}, \quad 0 < x < D_L.$$

The nontrivial conditions are

$$(x - a_i)^2 + b_i^2 \in \mathbb{Q}^\square,$$

where

$$\mathbb{Q}^\square = \{q^2 : q \in \mathbb{Q}\}$$

denotes the set of rational squares.

6.1 Pythagorean slopes and mirror classes

Definition 6.1 (Pythagorean-slope set). Define

$$\mathcal{S} = \{\alpha \in \mathbb{Q} : 1 + \alpha^2 \in \mathbb{Q}^\square\}.$$

Lemma 6.2 (Pythagorean-slope parametrization). *One has*

$$\mathcal{S} = \left\{ \frac{t - t^{-1}}{2} : t \in \mathbb{Q}^\times \right\},$$

and $\mathcal{S}$ is dense in $\mathbb{R}$.

Proof. If

$$\alpha = \frac{t - t^{-1}}{2},$$

then

$$1 + \alpha^2 = \frac{(t + t^{-1})^2}{4},$$

so $\alpha \in \mathcal{S}$. Conversely, if

$$\beta^2 = 1 + \alpha^2$$

with $\alpha, \beta \in \mathbb{Q}$, then

$$(\beta - \alpha)(\beta + \alpha) = 1.$$

Taking $t = \beta + \alpha$ gives

$$\alpha = (t - t^{-1})/2.$$

The real map $t \mapsto (t - t^{-1})/2$ is continuous and surjective from $\mathbb{R}^\times$ to $\mathbb{R}$. Since $\mathbb{Q}^\times$ is dense in $\mathbb{R}^\times$, the image $\mathcal{S}$ is dense in $\mathbb{R}$. $\square$

For an off-line point $Y_i = (a_i, b_i)$, set

$$\alpha_i(x) = \frac{x - a_i}{b_i}.$$

Then

$$PY_i \in \mathbb{Q} \iff \alpha_i(x) \in \mathcal{S}.$$

Indeed,

$$PY_i^2 = (x - a_i)^2 + b_i^2 = b_i^2 (1 + \alpha_i(x)^2),$$

and $b_i \in \mathbb{Q}^\times$.

Definition 6.3 (Mirror classes). Two off-line points $Y_i = (a_i, b_i)$ and $Y_j = (a_j, b_j)$ are mirror-equivalent with respect to L if

$$a_i = a_j, \quad b_i = \pm b_j.$$

Equivalently, they are equal or one is the reflection of the other across L . Thus they have equal distance from every point of L .

Lemma 6.4 (Mirror-class size). *With respect to a fixed line, a mirror class contains at most two distinct points. Therefore the three off-line vertices of a simple nondegenerate pentagon cannot all lie in one mirror class.*

Proof. In normalized coordinates, a mirror class is determined by the pair $(a, |b|)$ with $b \neq 0$. The only possible points in that class are (a, b) and $(a, -b)$. A pentagon has three distinct off-line vertices relative to a diagonal, so all three cannot lie in one class. $\square$

6.2 Genus and endpoint control

Lemma 6.5 (Genus from mirror classes). *Let $a_i, b_i \in \mathbb{Q}$ with $b_i \neq 0$. Remove duplicate equations coming from mirror-equivalent points in*

$$y_i^2 = (x - a_i)^2 + b_i^2.$$

If h mirror classes remain, then the smooth projective model has genus

$$g = 1 + 2^{h-1}(h - 2).$$

Thus

$$h = 1 \Rightarrow g = 0, \quad h = 2 \Rightarrow g = 1, \quad h = 3 \Rightarrow g = 5.$$

Proof. Over $\overline{\mathbb{Q}}$, the quadratic

$$f_i(x) = (x - a_i)^2 + b_i^2$$

has branch points

$$a_i \pm \sqrt{-1} b_i.$$

Two branch sets agree exactly when the two points are mirror-equivalent. After removing duplicates, the h branch sets are pairwise disjoint.

The square classes of the f_i are independent: any nonempty product has odd valuation at one branch point of any selected factor. Hence the cover has degree 2^h . It is branched over $2h$ finite points, and above each such point there are 2^{h-1} points with ramification index 2. There is no ramification above infinity because the f_i have degree 2. Riemann-Hurwitz gives

$$2g - 2 = -2 \cdot 2^h + (2h)2^{h-1} = 2^h(h - 2),$$

which is the formula. $\square$

Definition 6.6 (Endpoint control). Let L have endpoints U, V . A mirror class represented by Y is *endpoint-controlled* on L if

$$UY, VY \in \mathbb{Q}.$$

The definition is independent of the representative because reflection across L preserves distances from U and V .

Definition 6.7 (Arithmetic line type). Let h be the number of mirror classes of the three off-line vertices with respect to L . For a distinguished diagonal, call O the common endpoint and the other endpoint the far endpoint.

1. $h = 1$: genus 0. This is a formal easy line type, but it is vacuous for simple nondegenerate pentagons.
2. $h = 2$: genus 1. If both classes are endpoint-controlled, the line is endpoint-complete. If the common endpoint controls both classes but the far endpoint controls only one class, the line is a one-endpoint genus-one line.
3. $h = 3$: genus 5. If all three classes are endpoint-controlled, the line is endpoint-complete genus 5, called Type IIIa. If exactly one class is missing the far-endpoint distance while all classes are controlled at the common endpoint, the line is remote-vertex genus 5, called Type IIIb.

For the two distinguished pentagon diagonals, the common endpoint O controls every mirror class. On OX , the classes represented by A and C are endpoint-controlled because

$$OA, AX, OC, XC \in \mathbb{Z}.$$

The class represented by B is controlled at O , since $OB \in \mathbb{Z}$, but may be remote because XB is not assumed rational. On OC , the classes represented by X and B are endpoint-controlled because

$$OX, CX, OB, CB \in \mathbb{Z}.$$

The class represented by A is controlled at O , since $OA \in \mathbb{Z}$, but may be remote because AC is not assumed rational. Thus, for the distinguished diagonals in this problem, a two-class line which is not endpoint-complete is precisely a one-endpoint genus-one line.

6.3 Endpoint-complete genus one

Theorem 6.8 (Love detour density input). *Let*

$$s\alpha_1 + \alpha_2 = r$$

with $r, s \in \mathbb{Q}$, $s \neq 0$. Suppose

$$r \neq 0, \quad s \neq \pm 1, \quad 4r^2s \neq \pm(1 - s^2)^2.$$

On any real detour branch determined by fixed signs of

$$\sqrt{1 + \alpha_1^2}, \quad \sqrt{1 + \alpha_2^2},$$

the rational detour solutions are dense, provided that the branch contains one rational detour solution.

Proof. This is the detour-density form of Love's theorem used in this setting; see [11, Theorem 1.4 and Section 5]. Love attaches the elliptic curve

$$E_{r,s} : y^2 = x^3 + (1 + r^2 + s^2)x^2 + s^2x$$

and proves, outside the displayed exceptional locus, that the relevant detour point is non-torsion. Multiples of a non-torsion point are dense in the real elliptic component containing it. Pulling this density back through the detour parametrization gives density of rational detour solutions on the original real branch. $\square$

Theorem 6.9 (Two-point visible-interval theorem). *Let $U = (0, 0)$, $V = (D, 0)$, $D \in \mathbb{Q}_{>0}$, and let*

$$Y_1 = (a, b), \quad Y_2 = (c, d),$$

with $a, b, c, d \in \mathbb{Q}$ and $b > 0 > d$. Assume

$$UY_1, VY_1, UY_2, VY_2 \in \mathbb{Q}.$$

Put

$$s = \frac{b}{d}, \quad r = \frac{c-a}{d}.$$

If

$$r \neq 0, \quad s \neq \pm 1, \quad 4r^2s \neq \pm(1-s^2)^2,$$

then every nonempty open interval $J \subset (0, D)$ contains infinitely many rational points $P = (x, 0)$ such that

$$PY_1, PY_2 \in \mathbb{Q}.$$

Proof. For $P = (x, 0)$, set

$$\alpha_1 = \frac{x-a}{b}, \quad \alpha_2 = \frac{c-x}{d}.$$

Then

$$s\alpha_1 + \alpha_2 = r.$$

The distance conditions are exactly

$$\alpha_1, \alpha_2 \in \mathcal{S}.$$

For Y_2 , this uses

$$(x-c)^2 + d^2 = d^2(1 + \alpha_2^2),$$

because $(c-x)^2 = (x-c)^2$. The endpoint hypotheses give rational points on the relevant real detour branch above $x = 0$ and $x = D$, for instance on the branch with the positive choices of the two square roots. Theorem 6.8 makes rational detour solutions dense on that branch. Along that branch α_1 ranges over all real values, so

$$x = a + b\alpha_1$$

maps the branch onto $\mathbb{R}$. Hence the rational x -coordinates are dense in $(0, D)$, and therefore meet every nonempty open subinterval. $\square$

Theorem 6.10 (One-class line theorem). *Let L be a distinguished diagonal and let*

$$J \subset (\text{relint } L) \cap \mathcal{K}(Q)$$

be a nonempty open interval. If the three off-line vertices lie in one mirror class with respect to L , then J contains infinitely many rational points at rational distance from all five vertices.

Proof. Normalize L to the x -axis with endpoints $(0, 0)$ and $(D_L, 0)$. Under this normalization, J is identified with a nonempty open interval of x -values in $(0, D_L)$. Let the common mirror class be represented by $Y = (a, b)$, $b \neq 0$. A point $P = (x, 0)$ has rational distance from all off-line vertices exactly when

$$\frac{x-a}{b} \in \mathcal{S}.$$

The set $a + b\mathcal{S}$ is dense in $\mathbb{R}$ by Lemma 6.2; hence it meets J in infinitely many rational x -values. The endpoint distances are x and $D_L - x$, and the point lies in the kernel because $x \in J$. $\square$

Corollary 6.11 (One-class proof status). *If a distinguished diagonal has a nonempty open visible interval and the three off-line vertices determine one mirror class, then that diagonal is certified by infinitely many rational points. In a simple nondegenerate pentagon, this case cannot occur.*

Proof. The certification statement is Theorem 6.10. For the final assertion, Lemma 6.4 says that one mirror class contains at most two distinct off-line points, while a pentagon has three distinct off-line vertices relative to a diagonal. $\square$

Theorem 6.12 (Endpoint-complete nonexceptional two-class theorem). *Let L be a distinguished diagonal and let*

$$J \subset (\text{relint } L) \cap \mathcal{K}(Q)$$

be a nonempty open interval. Suppose the three off-line vertices determine exactly two mirror classes, both endpoint-controlled on L . Choose representatives Y_1, Y_2 on opposite sides of the normalized line, reflecting one representative across the line if necessary. Reflection does not change distances from points of L . Write

$$Y_1 = (a, b), \quad Y_2 = (c, d), \quad b > 0 > d,$$

and put

$$s = \frac{b}{d}, \quad r = \frac{c-a}{d}.$$

If

$$r \neq 0, \quad s \neq \pm 1, \quad 4r^2s \neq \pm(1-s^2)^2,$$

then J contains infinitely many rational points at rational distance from all five vertices.

Proof. After normalization, J becomes a nonempty open subinterval of $(0, D_L)$. Theorem 6.9 applies to Y_1 and Y_2 . Thus there are infinitely many rational points $P \in J$ with

$$PY_1, PY_2 \in \mathbb{Q}.$$

The third off-line vertex is mirror-equivalent to one of these representatives. The endpoint distances are rational for rational x on the diagonal. Since $J \subset \mathcal{K}(Q)$, the points are visible. $\square$

6.4 Exceptional genus one and genus five

Proposition 6.13 (Quartic model for two classes). *Let two non-mirror-equivalent off-line points on a normalized line be*

$$Y_1 = (a_1, b_1), \quad Y_2 = (a_2, b_2).$$

Assume $a_i, b_i \in \mathbb{Q}$ and $b_1b_2 \neq 0$. The two-distance curve

$$z_1^2 = 1 + \left(\frac{x - a_1}{b_1} \right)^2, \quad z_2^2 = 1 + \left(\frac{x - a_2}{b_2} \right)^2$$

is birational over $\mathbb{Q}$ to

$$W^2 = 4b_2^2T^2 + (b_1T^2 + 2(a_1 - a_2)T - b_1)^2.$$

Proof. Parametrize the first Pythagorean-slope condition by

$$\frac{x - a_1}{b_1} = \frac{T - T^{-1}}{2}.$$

Then

$$x = a_1 + \frac{b_1}{2}(T - T^{-1}).$$

Substitution into the second slope gives

$$\frac{x - a_2}{b_2} = \frac{b_1 T^2 + 2(a_1 - a_2)T - b_1}{2b_2 T}.$$

Multiplying the second square condition by $(2b_2 T)^2$ and setting

$$W = 2b_2 T z_2$$

gives the displayed quartic. The construction is reversible on dense affine opens, away from the finitely many points where $T = 0$ or one of the displayed rational transformations is undefined. $\square$

Definition 6.14 (Detour-exceptional two-class line). An endpoint-complete two-class line is called *detour-exceptional* if, after choosing representatives

$$Y_1 = (a, b), \quad Y_2 = (c, d)$$

on opposite sides with $b > 0 > d$, and writing

$$s = \frac{b}{d}, \quad r = \frac{c - a}{d},$$

at least one of

$$r = 0, \quad s = \pm 1, \quad 4r^2 s = \pm(1 - s^2)^2$$

holds. This condition is independent of which class is placed above the line: changing the choice replaces (r, s) by $(r/s, 1/s)$, and the displayed alternatives are unchanged.

Proposition 6.15 (Exceptional alternatives in nondegenerate pentagons). *For a nondegenerate pentagon whose three off-line vertices determine exactly two mirror classes with respect to L , choose representatives*

$$Y_1 = (a, b), \quad Y_2 = (c, d)$$

on opposite sides, $b > 0 > d$, and put

$$s = \frac{b}{d}, \quad r = \frac{c - a}{d}.$$

Then

$$s \neq 1, \quad r \neq 0.$$

Consequently the only endpoint-complete exceptional two-class alternatives that remain are

$$s = -1, \quad 4r^2 s = -(1 - s^2)^2.$$

Proof. Since $b > 0 > d$, one has $s < 0$, so $s = 1$ is impossible. If $r = 0$, then $a = c$. Every off-line vertex belongs to one of the two mirror classes, and reflection across the x -axis preserves the projection to the line. Hence all three off-line vertices have the same projection and are collinear on a line perpendicular to L , contradicting nondegeneracy. Finally, because $s < 0$, the quantity $4r^2s$ is negative, whereas $(1 - s^2)^2$ is nonnegative. Hence the positive-sign relation

$$4r^2s = (1 - s^2)^2$$

cannot occur. Among the two sign alternatives, only

$$4r^2s = -(1 - s^2)^2$$

can remain. □

Definition 6.16 (Hard arithmetic types). The arithmetic hard types are:

Type I. Endpoint-complete exceptional genus one. In the nondegenerate pentagon setting the remaining subtypes are

$$\text{I-a: } s = -1, \quad \text{I-b: } 4r^2s = -(1 - s^2)^2.$$

Type II. One-endpoint genus one.

Type III. Genuine genus 5. This splits into endpoint-complete genus 5, called Type IIIa, and remote-vertex genus 5, called Type IIIb.

Definition 6.17 (Refined hard subcases). The hard types are refined according to the arithmetic certificate available for their associated curve.

1. *Type I genus-one subcases.* For the exceptional endpoint-complete genus-one curve, the refined subcases are: rank-positive, rank-zero-hit, rank-zero-miss, and local-obstruction. These mean, respectively, that there is a non-torsion rational point on a real component projecting to a nonempty open subinterval of J_L ; that a complete finite torsion list contains a point mapping into J_L ; that the complete finite list has no point mapping into J_L ; or that there is a local obstruction compatible with J_L , or a local obstruction to the curve itself.
2. *Type II genus-one subcases.* The same four subcases are used for the one-endpoint genus-one curve.
3. *Type III genus-five subcases.* The refined subcases are: explicit point, elliptic quotient, genus-two quotient, full determination, local obstruction, and residual. These mean, respectively, that a rational point on the genus 5 curve maps into J_L ; that a genus-one quotient E_{ij} has been completely determined and its points have been checked for rational lifts to the full curve; that the genus 2 quotient H has been similarly determined and checked; that the full genus 5 rational point set has been determined; that a local obstruction rules out points compatible with J_L ; or that none of these computations is yet available.

Proposition 6.18 (Quotients in the genus 5 case). *Assume the three mirror classes on L are distinct, and write*

$$\alpha_i(x) = \frac{x - a_i}{b_i}$$

with $a_i, b_i \in \mathbb{Q}$ and $b_i \neq 0$. The curve

$$C_L : y_i^2 = 1 + \alpha_i(x)^2, \quad i = 1, 2, 3,$$

has three pairwise genus-one quotients

$$E_{ij} : y_i^2 = 1 + \alpha_i(x)^2, \quad y_j^2 = 1 + \alpha_j(x)^2,$$

and one genus 2 quotient

$$H : w^2 = \prod_{i=1}^3 (1 + \alpha_i(x)^2).$$

In the pentagon setting, the common endpoint O gives rational points on the E_{ij} , so these genus-one quotients may be regarded as elliptic curves after choosing one such point as the origin.

Proof. Forgetting one square root gives the two-distance curve E_{ij} . Since the two retained branch sets are distinct, Lemma 6.5 with $h = 2$ gives genus 1. Multiplying the three equations and setting $w = y_1 y_2 y_3$ gives the displayed hyperelliptic curve. The right side has the six distinct roots

$$a_i \pm \sqrt{-1} b_i, \quad i = 1, 2, 3,$$

so its smooth projective model has genus 2. Finally, in the pentagon setup the common endpoint O is at rational distance from each off-line vertex, as recorded above; hence the fibers of the E_{ij} above the common endpoint contain rational points. $\square$

7 Layer 3: certificates

The certificate layer is what prevents gaps. Geometry gives an interval, and arithmetic gives a curve. A positive certificate proves that the rational points of the curve actually hit the visible interval. A negative computation may also be useful, but only to rule out one diagonal before turning to the other distinguished diagonal.

Whenever an x -projection is mentioned below, the diagonal L has been put in the normalized coordinates of the arithmetic layer, and J_L is identified with its interval of x -values.

Definition 7.1 (Certificate types). Let L be a distinguished diagonal with visible interval J_L .

1. An *explicit point certificate* is a rational point $P \in J_L$ whose five vertex distances are directly verified to be rational.
2. An *exceptional interval certificate* for Type I is either a rational point of the exceptional two-class distance curve with $x \in J_L$, or a proof that such exceptional rational points are dense in a real interval I such that $I \cap J_L$ contains a nonempty open interval.
3. A *non-torsion interval certificate* for Type II consists of an elliptic model of the two-class curve, a non-torsion rational point R , and a proof that the real component containing R has x -projection whose intersection with J_L contains a nonempty open interval.

4. A *torsion certificate* is a complete rank-zero and torsion calculation whose finite list contains a point mapping into J_L .
5. A *genus 5 resolution certificate* is a rigorous determination of the relevant rational points and their x -coordinates, for example by an explicit point search, descent, Chabauty–Coleman, Mordell–Weil sieves, or a local obstruction. It is a positive certificate for L only when it produces an x -coordinate in J_L ; otherwise it is a negative certificate ruling out that diagonal.

Proposition 7.2 (Exceptional interval criterion). *In a detour-exceptional endpoint-complete two-class line problem, the diagonal L is certified if one has an exceptional interval certificate.*

Proof. A rational point of the exceptional two-class distance curve with $x \in J_L$ has rational distance from representatives of the two mirror classes; the third off-line distance is redundant, endpoint distances are rational, and membership in J_L gives visibility. If instead a dense exceptional family is proved on a real interval I with $I \cap J_L$ containing a nonempty open interval, then an exceptional rational point lies in that intersection, and the same distance and visibility argument applies. $\square$

Lemma 7.3 (Density on real elliptic components). *Let $E/\mathbb{Q}$ be an elliptic curve and $R \in E(\mathbb{Q})$ a non-torsion point. Then an appropriate congruence class of multiples of R is dense in the connected component of $E(\mathbb{R})$ containing R . If $E(\mathbb{R})$ is connected this congruence class is all multiples; if R lies outside the identity component when $E(\mathbb{R})$ has two components, it is the odd multiples.*

Proof. Let $E_0 = E(\mathbb{R})^0$ be the identity component. It is a compact connected one-dimensional real Lie group, hence isomorphic to $\mathbb{R}/\mathbb{Z}$. A non-torsion point in E_0 maps to an irrational rotation, so its cyclic subgroup is dense. If $R \in E_0$, this proves the claim. If $E(\mathbb{R})$ has two components and $R \notin E_0$, then $2R \in E_0$ is still non-torsion. The even multiples are dense in E_0 , and translating by R shows that the odd multiples are dense in the component containing R . $\square$

Proposition 7.4 (Certified one-endpoint criterion). *Let L be a distinguished diagonal and let*

$$J \subset (\text{relint } L) \cap \mathcal{K}(Q)$$

be a nonempty open interval. Use normalized coordinates on L , with the common endpoint O at $x = 0$. Suppose the off-line vertices reduce to two distinct mirror classes represented by

$$Y_1 = (a_1, b_1), \quad Y_2 = (a_2, b_2)$$

with $a_i, b_i \in \mathbb{Q}$ and $b_i \neq 0$, and suppose the common endpoint O is at rational distance from both representatives. Let

$$\mathcal{E} : \quad z_1^2 = 1 + \left(\frac{x - a_1}{b_1} \right)^2, \quad z_2^2 = 1 + \left(\frac{x - a_2}{b_2} \right)^2$$

and let $\tilde{\mathcal{E}}$ be its smooth projective model. If $\tilde{\mathcal{E}}$ has a non-torsion interval certificate for J , then J contains infinitely many rational points at rational distance from all five vertices.

Proof. The endpoint hypothesis gives a rational point of $\mathcal{E}$ above $x = 0$, because

$$1 + \left(\frac{-a_i}{b_i} \right)^2 = \frac{OY_i^2}{b_i^2}$$

is a rational square for $i = 1, 2$. Hence $\tilde{\mathcal{E}}(\mathbb{Q}) \neq \emptyset$. Since the two mirror classes are distinct, Lemma 6.5 gives genus 1, and any chosen rational point makes $\tilde{\mathcal{E}}$ an elliptic curve.

Let R be the non-torsion rational point supplied by the certificate. By Lemma 7.3, an appropriate congruence class of multiples of R is dense in the real component containing R . The certificate says that the x -projection of this component meets J in a nonempty open interval. The inverse image of that interval is a nonempty open subset of the component, so infinitely many rational multiples have x -coordinates in J .

For such a point, the defining equations give rational distances to the two mirror-class representatives. The mirror duplicate contributes no new distance condition. Finally, because $x \in J$ is rational in the normalized diagonal coordinate, the two endpoint distances are rational, and $J \subset \mathcal{K}(Q)$ gives visibility. $\square$

Proposition 7.5 (Refined genus-one hard-subcase criterion). *Let L be a Type I or Type II diagonal, and let E_L be the associated genus-one two-distance curve, with x -projection to the normalized diagonal. Let J_L be the visible interval.*

1. *In the rank-positive subcase, L is certified by infinitely many rational points.*
2. *In the rank-zero-hit subcase, L is certified by a finite calculation.*
3. *In the rank-zero-miss subcase, L is ruled out.*
4. *In the local-obstruction subcase, L is ruled out.*

Proof. In the rank-positive subcase, the hypothesis gives a non-torsion rational point R on a real component whose x -projection meets J_L in a nonempty open interval. Lemma 7.3 implies that an appropriate congruence class of multiples of R is dense in that component. Therefore infinitely many rational multiples have x -coordinates in J_L . The defining equations of the two-distance curve give rational distances to representatives of the two mirror classes; mirror-equivalent vertices add no new distance condition, and membership in $J_L \subset \mathcal{K}(Q)$ gives visibility.

In the rank-zero-hit subcase, the complete rank-zero and torsion calculation gives a finite list containing every rational point of E_L . If one listed point maps into J_L , its defining equations give the required rational off-line distances, the endpoint distances are rational in the normalized coordinate, and the point certifies L .

In the rank-zero-miss subcase, the same complete finite list contains every rational point of E_L , and none maps into J_L . Hence no rational point on this two-class curve can certify the diagonal. In the local-obstruction subcase, a rational point would give a point over every completion of $\mathbb{Q}$. The asserted local obstruction therefore forbids a rational point compatible with J_L , and the diagonal is ruled out. $\square$

Theorem 7.6 (Certified non-genus-5 line theorem). *Let L be a distinguished diagonal and let*

$$J \subset (\text{relint } L) \cap \mathcal{K}(Q)$$

be a nonempty open interval. Suppose the off-line vertices determine at most two mirror classes. Then L is certified in each of the following cases:

1. *one mirror class;*

2. two endpoint-controlled mirror classes with nonexceptional detour parameters;
3. two endpoint-controlled mirror classes with exceptional detour parameters and an exceptional interval certificate;
4. a one-endpoint genus-one line with a non-torsion interval certificate.

In cases 1, 2, and 4, there are infinitely many certified points in J .

Proof. In all four cases the hypothesis $J \subset \mathcal{K}(Q)$ converts a rational point of J with rational vertex distances into a certified point of the pentagon.

In case 1, Theorem 6.10 applies directly. The one mirror-class condition makes all three off-line distance conditions equivalent to one Pythagorean-slope condition. Since $\mathcal{S}$ is dense, there are infinitely many rational x -coordinates in J , and the endpoint distances are rational on the diagonal.

In case 2, Theorem 6.12 applies. Endpoint control gives rational detour points at the two endpoints, and the nonexceptional inequalities allow Love's density theorem to put infinitely many rational two-distance points into every nonempty open subinterval of $(0, D_L)$. Applying this to J gives infinitely many certified points.

In case 3, Proposition 7.2 applies. The exceptional interval certificate either gives an explicit rational point in J , or gives a dense exceptional family on an interval whose intersection with J contains a nonempty open interval. In either subcase one obtains a rational point of J at rational distance from representatives of both mirror classes, hence from all off-line vertices.

In case 4, Proposition 7.4 applies. The common endpoint supplies a rational point on the genus-one two-distance curve, the non-torsion interval certificate gives density on a real component whose x -projection meets J in a nonempty open interval, and hence infinitely many rational multiples map into J . $\square$

Theorem 7.7 (Certified nice-diagonal theorem). *Assume at least one distinguished diagonal is nice. Choose a nice distinguished diagonal L by the nice-first convention and normalize it as in Lemma 4.1. If the arithmetic line problem on L satisfies one of the certified non-genus-5 alternatives of Theorem 7.6, then Q is certified.*

Proof. Since L is nice, Proposition 5.6 gives the full visible interval

$$J_L = \text{relint } L.$$

This is a nonempty open interval. The arithmetic hypotheses place the line under Theorem 7.6, which supplies a rational point in J_L at rational distance from all five vertices. $\square$

Corollary 7.8 (Already proved and certificate-conditional branches). *Let L be a distinguished diagonal with a nonempty open visible interval*

$$J_L = (\text{relint } L) \cap \mathcal{K}(Q).$$

Then the following branches are either proved outright or proved once the stated positive certificate is supplied.

1. *If the off-line vertices determine one mirror class, then L is certified by infinitely many rational points. This branch is formal only: it is vacuous for simple nondegenerate pentagons.*

2. *If the off-line vertices determine exactly two endpoint-controlled mirror classes and the detour parameters are nonexceptional, then L is certified by infinitely many rational points.*
3. *If the line is Type I and has an exceptional interval certificate, then L is certified.*
4. *If the line is Type II and has a non-torsion interval certificate, then L is certified by infinitely many rational points.*
5. *If the line is Type III and has a positive genus 5 resolution certificate, then L is certified. A negative genus 5 resolution only rules out L .*

If L is nice, then $J_L = \text{relint } L$, so any one of the positive branches above certifies the pentagon Q . Niceness alone, however, supplies only the visibility interval and not the arithmetic certificate.

Proof. The first branch is Corollary 6.11. The second branch is Theorem 6.12. The Type I branch is Proposition 7.2. The Type II branch is Proposition 7.4. The Type III branch is exactly the positive-output clause in the genus 5 resolution program of Program 7.9: a computed rational x -coordinate in J_L gives a rational point on L with rational distance from all five vertices. If a genus 5 computation instead shows that no such x -coordinate lies in J_L , it proves only that this diagonal cannot certify the pentagon.

Finally, if L is nice, Proposition 5.6 gives

$$J_L = \text{relint } L.$$

The previous branches then produce a certified point on one of the distinguished diagonals, which is precisely the definition of certifying Q . Without one of those arithmetic or certificate inputs, niceness does not assert rational distances from the candidate point to the five vertices. $\square$

Program 7.9 (Genus 5 resolution program). For a Type III diagonal L , proceed as follows. A positive output certifies L . A negative complete output only rules out L ; the pentagon is then certified only if the other distinguished diagonal has a positive certificate.

1. Compute the visible interval J_L .
2. Write
$$C_L : \quad y_i^2 = 1 + \alpha_i(x)^2, \quad i = 1, 2, 3,$$
and verify that the six branch points are distinct.
3. Record whether the case is Type IIIa or Type IIIb.
4. Search for an explicit rational point with $x \in J_L$. Such a point immediately certifies the diagonal.
5. If no point is found, test local solubility subject to the rational inequalities defining J_L .
6. Compute the three elliptic quotients E_{ij} and the genus 2 quotient H of Proposition 6.18.
7. Use descent, Chabauty–Coleman, and Mordell–Weil sieves where their hypotheses are satisfied, recording the rank and saturation data needed for reproducibility.

8. Inspect the resulting rational x -coordinates. A coordinate in J_L certifies L ; a complete computation showing none lie in J_L rules out this diagonal and sends the proof to the other distinguished diagonal.

Proposition 7.10 (Refined Type III certificate criteria). *Let L be a Type III diagonal, and let*

$$C_L : y_i^2 = 1 + \alpha_i(x)^2, \quad i = 1, 2, 3,$$

be the associated genus 5 curve. Let E_{ij} and H be the quotients of Proposition 6.18.

1. *In the Type III-explicit subcase, L is certified.*
2. *Suppose a quotient E_{ij} has been completely determined and every rational point of E_{ij} whose x -coordinate lies in J_L has been checked for rational lifts to C_L . If one such lift exists, then L is certified; if none exists, then L is ruled out.*
3. *The same assertion holds with the genus 2 quotient H in place of E_{ij} .*
4. *If $C_L(\mathbb{Q})$ itself has been completely determined, then a point with $x \in J_L$ certifies L , while absence of such a point rules out L .*
5. *A Type III-local-obstruction subcase rules out L .*

Proof. In the explicit subcase, the displayed rational point on C_L gives rational values of all three square roots y_i , hence rational distances from the point $(x, 0)$ to all three off-line mirror-class representatives. The endpoint distances are rational because $x \in J_L$ is rational in the normalized diagonal coordinate, and $J_L \subset \mathcal{K}(Q)$ gives visibility.

For a quotient E_{ij} , every rational point of C_L maps to a rational point of E_{ij} . Therefore a complete determination of $E_{ij}(\mathbb{Q})$, followed by a rational-lift check above all quotient points with $x \in J_L$, is decisive. A lifted point certifies L by the preceding paragraph. If no such lift exists, then no rational point of C_L with $x \in J_L$ can exist, since any such point would have appeared in the checked quotient list. The proof for the genus 2 quotient H is identical: every rational point of C_L maps to a rational point of H , and the lift check recovers exactly the candidates on C_L .

If $C_L(\mathbb{Q})$ itself has been determined, the hit-or-miss conclusion is immediate by inspecting the finitely many x -coordinates. Finally, a rational point on C_L compatible with J_L would give a point over every completion of $\mathbb{Q}$. A local obstruction therefore rules out the diagonal. $\square$

8 The clean case division

The complete case division can now be written without mixing geometric, arithmetic, and certificate data. This section is a bookkeeping theorem: it proves that every distinguished diagonal lands in exactly one branch of the taxonomy, and it records what kind of positive certificate is needed to finish that branch.

Program 8.1 (Three-layer decision procedure). Given $Q = (O, A, X, C, B)$, do the following.

1. Verify the metric and geometric input: integral vertices, integral side lengths, integral diagonals OX, OC , simplicity, nondegeneracy, and interiority of the two diagonals.

2. Apply the nice-first convention. If a nice distinguished diagonal exists, choose it as the primary line OX . Otherwise choose either diagonal first, but compute both intervals if needed.
3. Geometry layer: for each diagonal under consideration, compute its visibility type by finding J_L . Use convexity or niceness as shortcuts when available. If J_L has no nonempty open interval, the interval density theorems do not apply on that diagonal, though an explicit point of J_L may still be checked directly.
4. Arithmetic layer: compute the mirror classes of the three off-line vertices. Record $h = 1, 2, 3$, endpoint status, and, when $h = 2$, the detour parameters r, s .
5. Certificate layer: apply the appropriate theorem or positive certificate. For $h = 1$, use Theorem 6.10. For endpoint-complete nonexceptional $h = 2$, use Theorem 6.12. For endpoint-complete exceptional $h = 2$, refine the case using Definition 6.17 and apply either Proposition 7.2 or Proposition 7.5. For one-endpoint $h = 2$, use the refined genus-one criterion. For $h = 3$, refine the case using Definition 6.17 and apply Proposition 7.10; a negative resolution only rules out that diagonal.
6. Stop as soon as one distinguished diagonal is certified. If the first diagonal is not certified, repeat the three layers on the other distinguished diagonal.

Theorem 8.2 (Three-layer certified case division). *For every pentagon in the setup, the three-layer decision procedure assigns to each distinguished diagonal exactly one geometric visibility type and exactly one arithmetic branch. The procedure certifies the pentagon exactly when at least one distinguished diagonal receives a valid positive certificate in the sense of the certificate layer.*

Proof. The geometry layer is exhaustive by Lemma 5.3: $\mathcal{K}(Q) \cap L$ is an interval, and intersecting it with relint L leaves an interval J_L , possibly empty. Therefore J_L is full, not full but containing a nonempty open interval, or degenerate.

For the arithmetic layer, the three off-line vertices determine $h = 1, 2$, or 3 mirror classes, and Lemma 6.5 assigns the corresponding genus. On a distinguished diagonal the common endpoint O controls every mirror class. Moreover, as recorded in the arithmetic layer, at most one mirror class can fail to be controlled by the far endpoint: on OX only the class represented by B can fail, and on OC only the class represented by A can fail. Hence the endpoint-complete and one-endpoint or remote-vertex alternatives in Definition 6.7 are exhaustive for $h = 2$ and $h = 3$. When $h = 2$ is endpoint-complete, the detour parameters either satisfy the nonexceptional inequalities or one of the exceptional equalities, so the nonexceptional and Type I exceptional branches are also exhaustive.

The certificate layer is a proof-status layer. Each positive certificate listed in the procedure is proved in the preceding theorems and propositions to imply a rational point in J_L at rational distance from all five vertices. Since $J_L \subset \mathcal{K}(Q)$, such a point certifies the diagonal and hence certifies the pentagon. Conversely, the procedure does not declare success unless one of the two distinguished diagonals has received such a positive certificate. A negative genus 5 resolution only rules out that diagonal and must be followed by the other diagonal. $\square$

Theorem 8.3 (Case-by-case provability audit). *Let L be a distinguished diagonal. The cases in the three-layer taxonomy have the following proof status in this paper.*

1. If J_L has no nonempty open interval, none of the interval-density theorems applies on L . Such a diagonal can still be certified by an explicit point certificate supported on J_L , if one exists.
2. If J_L has a nonempty open interval and $h = 1$, then L is certified by infinitely many rational points. For a simple nondegenerate pentagon this branch is vacuous.
3. If J_L has a nonempty open interval and $h = 2$ is endpoint-complete and nonexceptional, then L is uniformly certified by infinitely many rational points.
4. If $h = 2$ is endpoint-complete and detour-exceptional, then the only nondegenerate pentagon subtypes that remain are

$$s = -1, \quad 4r^2s = -(1 - s^2)^2.$$

These Type I branches are certified in this paper only after a positive Type I certificate is supplied. The refined subcases are the rank-positive, rank-zero-hit, rank-zero-miss, and local-obstruction subcases of Definition 6.17.

5. If $h = 2$ is one-endpoint genus one, then the branch is certified only after an explicit point certificate or a Type II non-torsion interval certificate is supplied. The same rank-positive, rank-zero-hit, rank-zero-miss, and local-obstruction refinement applies.
6. If $h = 3$, whether endpoint-complete Type IIIa or remote-vertex Type IIIb, then the branch is certified only after a positive genus 5 resolution certificate is supplied. A negative genus 5 resolution rules out this diagonal but does not certify the pentagon. The refined subcases are the explicit, elliptic-quotient, genus-two-quotient, full-determination, local-obstruction, and residual subcases of Definition 6.17.
7. Convexity, full visibility, and niceness are already proved geometry shortcuts. They certify Q only when combined with one of the positive arithmetic or certificate branches above.

Proof. The first item follows from the form of all density results used above: Theorems 6.10, 6.12, and Proposition 7.4 all require a nonempty open interval J_L . If J_L is empty, no point on L can certify the diagonal. If J_L is a single point or otherwise has no open subinterval, only direct verification of that point can certify the diagonal.

For $h = 1$, Corollary 6.11 gives infinitely many certified points whenever J_L has a nonempty open interval. The same corollary proves that the branch is vacuous in a simple nondegenerate pentagon, because one mirror class contains at most two distinct off-line vertices.

For endpoint-complete nonexceptional $h = 2$, the statement is precisely Theorem 6.12. The endpoint hypotheses give rational points on the relevant detour branch at the two endpoints, and the nonexceptional inequalities put the problem outside Love's exceptional locus. Love's density theorem then supplies infinitely many rational two-distance points in the visible interval.

For endpoint-complete exceptional $h = 2$, Proposition 6.15 eliminates $r = 0$, $s = 1$, and the positive-sign exceptional relation in the nondegenerate pentagon setting. Therefore only the displayed Type I subtypes remain. The nonexceptional density theorem deliberately excludes these subtypes, so the paper certifies them only through Proposition 7.2, or through an explicit point or complete torsion computation that is an explicit point certificate. These alternatives are formalized in Proposition 7.5.

For one-endpoint $h = 2$, the far endpoint does not control one of the two mirror classes. Hence the endpoint-complete detour theorem cannot be applied. Proposition 7.4 proves certification once a non-torsion interval certificate is supplied; an explicit point in J_L is also a certificate by definition. The rank-zero-hit, rank-zero-miss, and local-obstruction alternatives are again covered by Proposition 7.5.

For $h = 3$, Lemma 6.5 gives genus 5, and Proposition 6.18 records the lower-genus quotients available for computations. The paper does not assert a uniform density theorem for these genus 5 curves. Program 7.9 explains how a positive genus 5 resolution certifies L , and why a negative resolution only eliminates that particular diagonal. Proposition 7.10 proves the quotient, full-determination, explicit-point, and local-obstruction subcases.

Finally, Proposition 5.7 records exactly what convexity, kernel visibility, and niceness prove: they determine the visible interval. They do not impose rational-distance conditions. Thus they become full certifications only when one of the positive arithmetic or certificate branches above produces a rational point in that interval. $\square$

Theorem 8.4 (Finite and infinite branches). *For a distinguished diagonal L , define*

$$\mathcal{R}_L = \{P \in J_L : P \text{ is rational and } PV \in \mathbb{Q} \text{ for every vertex } V \text{ of } Q\}.$$

Then the following cardinality statements hold.

1. *If J_L has no nonempty open interval, then $\mathcal{R}_L$ is finite. In fact J_L is empty or a single point.*
2. *In the formal $h = 1$ branch, if J_L has a nonempty open interval, then $\mathcal{R}_L$ is infinite. This branch is vacuous for simple nondegenerate pentagons.*
3. *In the endpoint-complete nonexceptional $h = 2$ branch, if J_L has a nonempty open interval, then $\mathcal{R}_L$ is infinite.*
4. *In Type I exceptional genus one, the taxonomy alone does not decide whether $\mathcal{R}_L$ is finite or infinite. An exceptional interval certificate proves that $\mathcal{R}_L$ is infinite; an explicit point or rank-zero torsion certificate gives only a finite certified list.*
5. *In Type II one-endpoint genus one, the taxonomy alone does not decide whether $\mathcal{R}_L$ is finite or infinite. A non-torsion interval certificate proves that $\mathcal{R}_L$ is infinite; an explicit point or rank-zero torsion certificate gives only a finite certified list.*
6. *In Type III genus 5, the set $\mathcal{R}_L$ is finite. The difficulty is to determine this finite set, or at least determine whether one of its points has x -coordinate in J_L .*

Proof. The set J_L is an interval by Lemma 5.3. If an interval contains no nonempty open interval, then it is empty or a single point. Hence $\mathcal{R}_L$ has at most one point in that case.

The $h = 1$ assertion is Theorem 6.10, and the vacuity assertion is Corollary 6.11. The endpoint-complete nonexceptional $h = 2$ assertion is Theorem 6.12. Both results produce infinitely many rational points in every nonempty open visible interval.

For Type I, an exceptional interval certificate gives a dense exceptional family on a real interval meeting J_L in a nonempty open interval. Therefore infinitely many rational points of the exceptional two-class distance curve map into J_L . By contrast, an explicit point certificate proves only the listed point, and a rank-zero torsion certificate proves membership by checking a finite list of rational

points. The same dichotomy holds for Type II: Proposition 7.4 proves infinitude from a non-torsion interval certificate, while an explicit point or rank-zero torsion calculation supplies only finitely many certified candidates.

Finally suppose $h = 3$. Lemma 6.5 gives a smooth projective model of genus 5 for the full three-distance curve. By Faltings' theorem [6], this curve has only finitely many rational points. The set $\mathcal{R}_L$ is the image, under the x -coordinate map, of a subset of those rational points satisfying the visible-interval condition; therefore $\mathcal{R}_L$ is finite. $\square$

Example 8.5 (A degenerate finite diagonal). Let

$$O = (0, 0), \quad A = (3, -4), \quad X = (3, 0), \quad C = (0, 4), \quad B = (-12, -5),$$

and let $Q = (O, A, X, C, B)$. The side lengths are

$$OA = 5, \quad AX = 4, \quad XC = 5, \quad CB = 15, \quad BO = 13,$$

and the distinguished diagonals have lengths

$$OX = 3, \quad OC = 4.$$

The polygon is simple and nondegenerate, and the two distinguished diagonals are interior diagonals.

We compute the visible interval on OX . Write a point of OX as

$$P = (t, 0), \quad 0 \leq t \leq 3.$$

The vertices are counterclockwise, so the kernel is cut out by the left half-planes of the five oriented sides. Substituting $P = (t, 0)$ into these five inequalities gives

$$t \geq 0, \quad t \leq 3, \quad t \leq 3, \quad 48 + 9t \geq 0, \quad -5t \geq 0.$$

Thus

$$\mathcal{K}(Q) \cap OX = \{O\}.$$

Consequently

$$J_X = (\text{relint } OX) \cap \mathcal{K}(Q) = \emptyset,$$

and hence

$$\mathcal{R}_X = \emptyset.$$

This is a full pentagon example of the finite degenerate-visibility case: the finite set is empty on the diagonal OX .

Example 8.6 (Finite Type IIIa and Type IIIb diagonals). First let

$$Q_a = (O, A, X, C, B)$$

with

$$O = (0, 0), \quad A = (3, 4), \quad X = (6, 0), \quad C = (6, -8), \quad B = (0, -8).$$

The side lengths are

$$OA = 5, \quad AX = 5, \quad XC = 8, \quad CB = 6, \quad BO = 8,$$

and

$$OX = 6, \quad OC = 10.$$

The polygon is strictly convex, so $J_X = (0, 6)$. With respect to OX , the three off-line vertices have normalized coordinates

$$A = (3, 4), \quad C = (6, -8), \quad B = (0, -8).$$

Their mirror classes are distinct:

$$(3, 4), \quad (6, 8), \quad (0, 8).$$

Moreover all three classes are endpoint-controlled on OX , since

$$XA = 5, \quad XC = 8, \quad XB = 10.$$

Thus OX is a Type IIIa diagonal. By Theorem 8.4, the set $\mathcal{R}_X$ is finite.

For a remote-vertex example, let

$$Q_b = (O, A, X, C, B)$$

with

$$O = (0, 0), \quad A = (3, 4), \quad X = (6, 0), \quad C = (6, -8), \quad B = (0, -16).$$

The side lengths are

$$OA = 5, \quad AX = 5, \quad XC = 8, \quad CB = 10, \quad BO = 16,$$

and again

$$OX = 6, \quad OC = 10.$$

This polygon is also strictly convex, so $J_X = (0, 6)$. The mirror classes with respect to OX are

$$(3, 4), \quad (6, 8), \quad (0, 16),$$

so $h = 3$. The classes represented by A and C are endpoint-controlled on OX , while the class represented by B is not controlled at the far endpoint because

$$XB^2 = 6^2 + 16^2 = 292$$

is not a rational square. Thus OX is a Type IIIb diagonal. Again Theorem 8.4 shows that $\mathcal{R}_X$ is finite.

Remark 8.7 (Finite certificates versus finite solution sets). An explicit point certificate gives a finite certified list only because it lists finitely many checked points; it does not prove that no other solutions exist. Similarly, a rank-zero torsion certificate is a finite certificate only after a complete rank and torsion computation has been supplied. The two examples above illustrate the finite cases that are proved uniformly by the present paper: degenerate visibility and Type III genus 5.

9 How this differs from convex, kernel-shape, and niceness taxonomies

The three-layer case division clarifies the role of several more familiar taxonomies.

1. *Convexity is geometric.* Convexity proves $\mathcal{K}(Q) = Q$, hence gives full visibility on every interior diagonal. It does not determine the mirror classes, endpoint-control data, detour parameters, or genus of the associated line curve. Thus, after convexity has supplied full visibility, the arithmetic branch still has to be computed.
2. *Kernel shape is auxiliary.* Once the two intervals J_X and J_C have been computed, the arithmetic layer uses only these one-dimensional intervals. The full polygonal shape of $\mathcal{K}(Q)$ remains useful for computing or checking the intervals, but it is not itself the arithmetic case split.
3. *Niceness is a visibility shortcut.* If L is nice, then

$$J_L = \text{relint } L$$

by Proposition 5.6. This justifies the nice-first normalization convention when a nice distinguished diagonal is available. After that geometric simplification, the mirror-class, endpoint-control, detour, genus, and certificate questions still remain.

4. *The diagonal profile is the basic object.* The systematic classification is attached to a pair (Q, L) , not only to Q . It records

J_L , mirror classes, endpoint-control data,
 detour or genus branch,
 positive certificate or negative resolution status.

The pentagon is certified only when one of the two distinguished diagonal profiles has a positive certificate.

10 Conclusion

The reorganized proof has three separate steps. The geometry layer computes the visible intervals J_X and J_C . The arithmetic layer classifies the corresponding line problems by mirror classes, endpoint-control data, detour parameters, and genus. The certificate layer then supplies either a positive certificate, which certifies a diagonal, or a negative resolution, which can only rule out that diagonal.

This separation is the main point of the taxonomy. Convexity, full-kernel visibility, and niceness settle the geometry layer, but they do not by themselves settle the arithmetic or certificate layers. Conversely, an arithmetic theorem is useful for the pentagon only after its rational points are shown to meet the visible interval on one of the two distinguished diagonals.

Thus the remaining uniform proof problem is precise: for every pentagon in the stated class, one must produce a positive certificate on at least one of the two distinguished diagonal profiles. The hard work is concentrated in the Type I, Type II, and Type III branches, where exceptional genus-one, one-endpoint genus-one, and genuine genus 5 phenomena must be resolved with certificates that actually hit the corresponding visible interval.

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